

Researchify Studio

Minor and Major Project Functional Division

Project Overview

Researchify Studio is an AI-assisted research platform designed to simplify and improve the academic research workflow for students, researchers, scholars, and professionals. Academic research generally requires users to move between multiple platforms for discovering papers, downloading and organizing documents, reading lengthy research articles, maintaining notes, generating citations, comparing studies, and identifying relevant research directions. Researchify Studio aims to bring these activities into a unified research environment.

The project is divided into two progressive academic stages: a **Minor Project** and a **Major Project**. The Minor Project will establish the core research discovery, paper management, document processing, and basic AI capabilities of the platform. The Major Project will extend this foundation into an advanced AI research intelligence system using **Retrieval-Augmented Generation (RAG), embeddings, vector databases, semantic retrieval, multi-document analysis, citation-aware responses, and research knowledge management**.

The division is intentionally progressive. The Minor Project will be a complete and independently demonstrable application rather than an incomplete prototype. The Major Project will reuse and extend the same architecture, database, user system, research library, and document-processing infrastructure while introducing significantly more advanced artificial intelligence capabilities.

Part I — Minor Project

Project Title

Researchify Studio: AI-Assisted Research Paper Discovery and Management Platform

Project Objective

The objective of the Minor Project is to develop a centralized research platform that assists users in **discovering, exploring, organizing, uploading, managing, and understanding academic research papers**.

The system will integrate academic research sources with a personalized research library and basic artificial intelligence capabilities. Users will be able to search for scholarly papers, examine their metadata and abstracts, save relevant papers into collections, upload their own PDF documents, maintain research notes, and obtain AI-generated summaries.

The Minor Project will primarily demonstrate knowledge of **full-stack web development, REST APIs, database management, authentication, third-party academic API integration, PDF processing, and basic Large Language Model integration.**

Its fundamental workflow can be represented as:

Discover → Explore → Save → Organize → Upload → Summarize → Manage

1. User Authentication and Research Profile

The platform will provide a secure authentication system through which researchers can create and manage individual accounts. Registration, login, logout, protected routes, authentication tokens, and user-session management will form the foundation of personalized functionality.

Each user will maintain a research profile containing basic information such as name, email address, academic field, research interests, and other relevant preferences. These interests may later be used to personalize paper recommendations and the research dashboard.

This module will ensure that saved papers, uploaded documents, notes, collections, search history, and other research activities remain associated with the appropriate user.

2. Personalized Research Dashboard

After authentication, users will be provided with a centralized research dashboard representing their current research activity.

The dashboard will provide quick access to important information such as saved papers, uploaded documents, research collections, recent searches, recently accessed papers, notes, and recommended academic content.

Rather than functioning merely as a homepage, the dashboard will act as the user's primary research control center. It will allow researchers to quickly continue previous work and access the major features of the application.

3. Academic Research Paper Discovery

Research paper discovery will be one of the primary modules of the Minor Project. The platform will integrate suitable scholarly-data APIs or open academic databases to retrieve research-paper information.

Users will be able to search using research topics, keywords, paper titles, authors, or related terms. Search results may contain information such as the paper title, authors, abstract, publication year, journal or conference, DOI, citation information where available, and links to the original publication or openly available document.

The objective is to reduce dependence on manually navigating multiple research websites by providing a centralized discovery interface.

4. Search Filtering and Sorting

To improve research discovery, the system will provide filtering and sorting capabilities.

Researchers may refine results according to parameters such as publication year, author, research topic, relevance, publication recency, citation count where supported by the underlying source, and open-access availability.

This module will demonstrate practical implementation of API query parameters, dynamic filtering, state management, and user-focused search interfaces.

5. Research Paper Details

Every selected research paper will have a dedicated details view containing the academic information available for that publication.

The page may display the title, author information, abstract, publication year, journal or conference, DOI, keywords, citation information, related papers, and available external or PDF links.

Users will also be able to save papers directly to their personal research library or assign them to an appropriate collection.

6. Personal Research Library

The Research Library will provide users with a permanent workspace for storing relevant academic literature.

Instead of repeatedly searching for the same research papers, users will be able to bookmark and organize important papers inside Researchify Studio. The library will support operations such as saving papers, removing papers, marking important papers as favorites, searching saved papers, and organizing them into research collections.

For example, a researcher working on artificial intelligence may create separate collections for **Retrieval-Augmented Generation, Large Language Models, Computer Vision, Natural Language Processing, and Research Methodology**.

The library created during the Minor Project will later become an important knowledge source for the Major Project's AI research assistant.

7. Research Collections

Users will be able to create customized collections for organizing academic papers according to projects, subjects, topics, or research objectives.

A researcher could, for example, create a project-specific collection and add multiple discovered or uploaded papers to it. Collections will provide better organization than a simple bookmark system and will establish the foundation for multi-paper research workspaces in the Major Project.

Basic collection operations will include creation, modification, deletion, adding papers, removing papers, and viewing papers associated with a particular collection.

8. Research Paper PDF Upload and Processing

In addition to discovering papers through academic sources, users will be able to upload research papers in PDF format.

The system will process uploaded documents and extract readable textual content and available metadata. Basic information such as document name, title where detectable, number of pages, upload date, and extracted text may be maintained.

The Minor Project will focus primarily on reliable PDF upload and text extraction rather than advanced document intelligence.

The processing pipeline will broadly follow:

PDF Upload → Document Validation → Text Extraction → Metadata Processing → Storage → AI Processing

This infrastructure will later become the foundation of the Major Project's RAG pipeline.

9. AI-Assisted Research Paper Summarization

AI-based paper summarization will represent the primary artificial intelligence feature of the Minor Project.

Users will be able to select an uploaded or supported research paper and request an automatically generated summary. Instead of only producing an unrestricted paragraph, Researchify Studio can organize the output into useful academic sections such as:

Quick Summary, Research Objective, Methodology, Key Findings, Results, Limitations, and Conclusion, depending on information available in the paper.

This feature will introduce Large Language Model integration while remaining technically manageable within the scope of a Minor Project.

The simplified architecture will be:

Research Paper → Extracted Text → LLM Processing → Structured Summary

10. Research Notes

Researchers will be provided with a basic note-management system.

Users will be able to create, edit, save, and delete notes and associate those notes with specific research papers or collections. This allows important observations, findings, ideas, questions, or references to remain connected with the relevant academic material.

The Minor version will primarily focus on manual note management. AI-assisted note organization and knowledge extraction will be introduced during the Major Project.

11. Basic Research Paper Recommendations

The Minor Project may include a basic recommendation system based on readily available information such as the user's research interests, previous searches, saved-paper keywords, selected categories, or frequently accessed topics.

For example, if a researcher frequently searches for and saves papers related to RAG and LLMs, the platform may prioritize similar academic papers.

The objective at this stage is not to develop a sophisticated machine-learning recommendation engine. Instead, it establishes the personalization infrastructure that can later be upgraded using embeddings and semantic similarity.

Minor Project Scope Summary

At completion, the Minor Project should provide the following major functional modules:

Module	Minor Project Functionality
Authentication	Registration, login, logout, protected user accounts
Research Profile	Academic information and research interests
Dashboard	Personalized research overview
Paper Discovery	Search academic papers using external scholarly sources
Search Filters	Year, relevance, author, recency and available filters
Paper Details	Metadata, abstract, authors, DOI and related information
Research Library	Save, remove, favorite and manage papers
Collections	Organize papers by topic/project
PDF Processing	Upload PDF and extract text/metadata
AI Summarization	Generate structured paper summaries
Research Notes	Create and manage paper-related notes
Recommendations	Basic personalized paper suggestions

Minor Project Expected Outcome

At the end of the Minor Project, Researchify Studio will function as an **AI-assisted academic paper discovery and research management platform**.

A user should be able to create an account, discover academic papers, filter search results, examine publications, maintain a personal research library, organize papers into

collections, upload PDF documents, generate AI-assisted summaries, and maintain research notes.

Therefore, the Minor Project will itself represent a complete usable system while establishing the technical foundation required for the Major Project.

Part II — Major Project

Project Title

Researchify Studio: RAG-Based AI Research Intelligence and Knowledge Workspace

Project Objective

The Major Project will transform the research management platform developed during the Minor Project into an advanced **AI-powered research intelligence and knowledge workspace**.

Rather than simply helping users locate and summarize research papers, the Major version will allow researchers to interact intelligently with individual documents, multiple papers, collections, and their broader research workspace.

The central technological advancement will be the implementation of **Retrieval-Augmented Generation (RAG)** combined with document embeddings, vector databases, semantic search, contextual retrieval, Large Language Models, and citation-aware generation.

The Major Project workflow will therefore evolve into:

Discover → Organize → Understand → Ask → Retrieve → Compare → Analyze → Synthesize → Cite → Generate Research Insights

1. Retrieval-Augmented Generation Architecture

RAG will form the technical core of the Major Project.

Instead of sending entire research documents directly to an LLM, uploaded papers will undergo a structured processing pipeline. Extracted text will be cleaned and divided into meaningful chunks. Embedding models will convert these chunks into numerical vector representations, which will then be stored inside a vector database.

When the researcher asks a question, Researchify Studio will search the vector database for the most semantically relevant sections and provide those sections as context to the language model.

The architecture will broadly follow:

PDF → Text Extraction → Cleaning → Chunking → Embeddings → Vector Database → Semantic Retrieval → Relevant Context → LLM → Evidence-Grounded Answer

This approach improves scalability and allows answers to be grounded in the researcher's actual documents.

2. Chat with a Research Paper

Users will be able to interact conversationally with an uploaded research paper.

Instead of reading an entire document to locate a particular piece of information, the researcher may ask questions such as:

| What methodology does this paper use?

| What dataset was used for evaluation?

| What limitations did the authors identify?

| What are the main experimental findings?

The RAG pipeline will retrieve the most relevant document sections before generating an answer.

Where technically feasible, answers should provide references to the corresponding paper, section, page, or retrieved source passage. This will differentiate Researchify Studio from a generic chatbot.

3. Multi-Paper Research Chat

The Major Project will extend document conversation beyond a single paper.

Researchers will be able to select several papers from their library or research project and ask questions across the entire selection.

For example:

- Compare the methodologies used by these five papers.
- Which papers report limitations related to dataset size?
- What approaches are commonly used across these studies?
- Which studies produced contradictory findings?

The system will retrieve relevant information across multiple documents before constructing a combined response.

This capability will make Researchify Studio useful for actual literature analysis rather than simple document summarization.

4. AI-Based Research Paper Comparison

The platform will provide a dedicated paper-comparison feature.

Users may select multiple research papers and request comparison across academic criteria such as:

- Research objective
- Problem addressed
- Methodology
- Dataset
- Models or algorithms
- Experimental setup
- Results
- Advantages
- Limitations
- Future work

The resulting information can be displayed in structured comparison tables accompanied by an AI-generated comparative analysis.

This module will help researchers quickly understand differences and similarities among related studies.

5. AI Literature Review Assistance

A major feature of the advanced system will be AI-assisted literature review generation.

Researchers will select a set of relevant papers, after which Researchify Studio will retrieve and synthesize information across the selected sources.

The generated review may be structured into sections such as **Introduction, Existing Approaches, Major Themes, Comparative Findings, Limitations, Research Trends, Research Gaps, and Conclusion.**

The objective should be to assist researchers in synthesizing literature rather than automatically replacing scholarly judgment. Wherever possible, generated statements should remain traceable to the papers from which supporting information was retrieved.

6. Research Gap Identification

The system will analyze selected research literature to identify potential underexplored areas.

Researchify Studio may examine information such as limitations reported by authors, proposed future work, dataset restrictions, methodological weaknesses, conflicting findings, underrepresented populations or domains, and areas receiving comparatively little investigation.

The system can then present **potential research gaps accompanied by supporting evidence from the analyzed papers.**

This feature is particularly valuable for students beginning dissertations, final-year projects, research proposals, or academic publications.

7. Semantic Research Search

The Major Project will extend conventional keyword search using semantic retrieval.

Traditional keyword search primarily depends on matching words appearing in the query and document. Semantic search uses embeddings to represent the meaning of queries and document content.

For example, a researcher searching for:

"methods for reducing incorrect facts generated by language models"

may retrieve literature discussing **hallucination mitigation**, **factual grounding**, **knowledge augmentation**, **retrieval-based generation**, or **factual consistency**, even when the paper does not contain the exact wording of the original query.

Semantic retrieval can be applied to uploaded documents, saved research libraries, collections, or project workspaces.

8. Global AI Research Assistant

The Major Project will introduce an intelligent research assistant capable of operating over the user's research environment.

Instead of being restricted to a single PDF, the assistant can potentially understand the user's selected research project, saved papers, uploaded documents, collections, and research notes.

Researchers may ask questions such as:

Which papers in my library discuss hallucination mitigation?

Compare the RAG architectures used in my selected papers.

Summarize the major findings across this project.

Which papers disagree with each other?

What limitations repeatedly appear in this collection?

The assistant therefore becomes a contextual interface to the user's research knowledge base.

9. Advanced Research Projects and Workspaces

The basic collections introduced during the Minor Project will evolve into complete research workspaces.

Each research project may contain:

Papers, Uploaded Documents, Collections, Notes, AI Conversations, Literature Reviews, Paper Comparisons, Research Gaps, References, and Generated Reports.

For example, a researcher studying *LLM Hallucination Reduction* could maintain all related literature, analysis, notes, AI conversations, and citations inside a single project.

This feature changes Researchify Studio from a paper-management application into a persistent research environment.

10. Citation and Reference Management

The Major Project will introduce citation-management functionality.

Using available paper metadata, users will be able to generate references in commonly used academic citation styles such as:

- APA
- IEEE
- MLA
- Chicago
- Harvard

Where appropriate, references may also be exported in formats such as **BibTeX** or **RIS**.

This functionality can be integrated with the research library, literature-review module, notes, and generated research reports.

11. Citation-Aware and Evidence-Grounded AI Responses

An important objective of the Major Project should be improving the traceability of AI-generated information.

When the research assistant makes claims based on selected papers, the system should associate those claims with the supporting sources whenever technically feasible.

For example, rather than returning an unsupported AI response, the interface may provide numbered references to the research papers and relevant retrieved sections used in producing the response.

This increases academic usefulness and helps researchers independently verify generated information.

12. Advanced Document Intelligence

The PDF-processing infrastructure developed in the Minor Project will be expanded substantially.

The advanced pipeline may include:

Document Upload → PDF Parsing → Metadata Extraction → Section Identification → Reference Extraction → Text Cleaning → Chunking → Embedding Generation → Vector Storage → AI Analysis

Depending on implementation feasibility, the system may additionally identify structured elements such as abstracts, methodologies, conclusions, references, tables, and figures.

The purpose of this module is to transform uploaded PDFs from static files into searchable and machine-interpretable research knowledge.

13. AI-Assisted Research Notes

The manual notes system developed during the Minor Project will be enhanced with artificial intelligence.

Researchers may use AI to summarize selected information, explain difficult concepts, reorganize notes, extract important points, identify related papers, categorize observations, or connect notes with existing research material.

Over time, these notes can become part of the researcher's broader project knowledge base and may be made available to the contextual research assistant.

14. Advanced Research Analytics

The Major Project will provide analytics describing the user's research activity and knowledge base.

Potential analytics include the number of papers discovered, papers saved, documents analyzed, frequently researched topics, most referenced authors, publication-year distributions, research activity over time, collection statistics, AI-query activity, and project-level research progress.

These analytics will provide users with a visual overview of their research behavior and accumulated academic resources.

15. Research Output and Export

The completed platform should allow users to export selected research outputs for further academic work.

Depending on the content, users may export:

- Research notes
- Paper comparisons
- Literature-review drafts
- Bibliographies
- Research reports
- Research-gap analyses
- Selected research summaries

Appropriate output formats may include **PDF, DOCX, Markdown, BibTeX, or RIS**.

This ensures that research conducted inside Researchify Studio can be transferred into external academic workflows.

Major Project Scope Summary

Module	Major Project Functionality
Minor Project Features	Retained and improved
RAG Pipeline	Document chunking, embeddings, retrieval and grounded generation
Vector Database	Semantic storage and retrieval
Chat with PDF	Contextual question answering over individual papers
Multi-Paper Chat	Questions and analysis across selected literature
Paper Comparison	Structured AI comparison of multiple studies
Literature Review	Multi-source literature synthesis
Research Gap Analysis	Evidence-supported potential gap identification
Semantic Search	Meaning-based research retrieval

Module	Major Project Functionality
AI Research Assistant	Contextual assistant over research workspace
Research Workspace	Project-based knowledge organization
Citation Management	Academic citation generation and export
Evidence-Grounded AI	Source-linked AI responses
Document Intelligence	Advanced parsing and document processing
AI Notes	AI-assisted research note management
Analytics	Advanced research activity and knowledge analytics
Export	Research reports, reviews, citations and notes

Minor-to-Major Technical Progression

The two stages demonstrate a clear increase in technical complexity.

Technical Area	Minor Project	Major Project
Full-Stack Development	Core	Advanced
Authentication	Yes	Extended
Database	MongoDB/application data	Extended research knowledge data
Academic APIs	Paper discovery	Discovery + intelligence
PDF Processing	Text extraction	Advanced document processing
Artificial Intelligence	Summarization	Contextual research intelligence
LLM Integration	Basic	Advanced
Python	Basic/Optional	Significant
RAG	No	Core technology
Embeddings	No	Yes

Technical Area	Minor Project	Major Project
Vector Database	No	Yes
Semantic Retrieval	No	Yes
Multi-Document Analysis	No	Yes
Citation-Aware Generation	No	Yes
Research Knowledge Base	Basic library	Advanced contextual workspace

Overall Project Progression

The academic progression of Researchify Studio can therefore be summarized as follows:

Minor Project — Research Discovery and Management

The Minor Project focuses on building the **foundation of the research ecosystem**. It solves the problems of fragmented paper discovery, research organization, document management, and time-consuming initial paper understanding.

Its central capability is:

Finding and managing research information efficiently.

Major Project — Research Intelligence and Knowledge Assistance

The Major Project builds upon that foundation and addresses the more complex problem of **understanding relationships across large amounts of research literature**.

Instead of merely storing and summarizing papers, the system retrieves evidence, answers questions from documents, compares studies, assists with literature synthesis, identifies potential gaps, manages references, and builds a contextual research knowledge workspace.

Its central capability becomes:

Understanding, connecting, analyzing, and synthesizing research information intelligently.

Final Academic Positioning

The **Minor Project** can formally be described as:

Researchify Studio is an AI-assisted research paper discovery and management platform designed to centralize academic literature search, research-paper organization, PDF processing, research notes, personalized libraries, and AI-assisted document summarization. The system aims to reduce the fragmentation and time involved in the initial stages of academic research while providing researchers with a structured digital workspace for managing scholarly information.

The **Major Project** can formally be described as:

Researchify Studio is a RAG-based AI research intelligence and knowledge workspace that extends conventional research management through semantic retrieval, vector-based document indexing, evidence-grounded question answering, multi-paper analysis, literature synthesis, citation management, research-gap identification, and contextual AI assistance. The system aims to provide researchers with an integrated environment for discovering, organizing, understanding, comparing, and synthesizing scholarly knowledge while maintaining traceability to the underlying academic sources.

The important architectural principle is that these are **not two unrelated projects**. The Minor Project establishes the production foundation—users, search, papers, collections, PDFs, notes, database, dashboard and basic AI—while the Major Project upgrades that same system with **RAG, embeddings, vector retrieval, advanced document intelligence and multi-document research reasoning**. This makes the progression technically defensible for academic evaluation while ensuring that almost everything developed during the Minor Project remains useful in the final Major Project.